Apr. 7

- Reid Lecture next Tuesday right after our class
- Exam 2 this Thursday, content up to 1st Isomorphism Theorem

- **The first Isomorphism Theorem**: Let $\phi: G \to H$ be a homomorphism. Then $G/\ker\phi \cong \phi(G)$
  - $\ker\phi = \{a \in G : \phi(a) = e_H\} \trianglelefteq G$
  - $\phi(G) = \operatorname{im}(G) = \{b \in H : \exists a \in G \text{ with } \phi(a) = b\} \leq H$
  - $G/\ker\phi = \{a\ker\phi : a \in G\}$

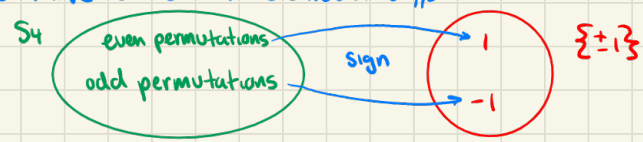
everything in $\ker\phi$ "becomes the identity"
e.g mod 5, multiples of 5 "become" 0

Ex. $\operatorname{sign}: S_4 \to \{1,-1\}$ given by $\operatorname{sgn}(\alpha) = \begin{cases} 1 & \text{if } \alpha \text{ is even} \\ -1 & \text{if } \alpha \text{ is odd} \end{cases}$

Can show this is homomorphism, 4 cases

- $\operatorname{sgn}(\alpha\beta) = 1$ if α, β both even, $\operatorname{sgn}(\alpha)\operatorname{sgn}(\beta) = 1 \cdot 1 = 1$

  3 more cases with even/odd α, β



$S_4/\ker\operatorname{sign} \cong \operatorname{sgn}(S_4) = \{1,-1\}$
$\quad\quad\quad

- <u>The Third Isomorphism Theorem</u> Suppose $M, N \trianglelefteq G$ and $N \leq M$. Then $(G/N)/(M/N) \cong G/M$

since $N \trianglelefteq G$, $N \trianglelefteq M$

$N \trianglelefteq M \trianglelefteq G \longrightarrow \underset{\{mN,\, m\in M\}}{\underset{\shortparallel}{M/N}} \triangleleft \underset{\{gN:\, g\in G\}}{\underset{\shortparallel}{G/N}}$

$gNmNg^{-1}N = gmg^{-1}N = m'N$

Criterion for normality

Skipped proving it's a homomorphism for time

<u>Proof</u> Define $\Phi: G/N \to G/M$ by $\Phi(gN) = gM$. By the FIT, $(G/N)/\ker\Phi \cong \Phi(G/N)$.

To show $\Phi(G/N) = G/M$, let $gM \in G/M$. We see $\Phi(gN) = gM$, so Φ is surjective. So, we have $(G/N)/\ker\Phi \cong G/M$. It remains to show $\ker\Phi = M/N$.

$\ker\Phi = \{gN : \Phi(gN) = eM\} = \{gN : \Phi(gN) = gM = eM\} = \{gN : g \in M\} = M/N$.

Therefore, $(G/N)/(M/N) \cong G/M$. $\square$

- Showing $\Phi: G/N \to G/M$ by $\Phi(gN) = gM$ is well-defined.

Suppose $gN = g'N$. We want $\Phi(gN) = \Phi(g'N)$. Observe $\Phi(gN) = gM$ and $\Phi(g'N) = g'M$. Since $gN = g'N \Leftrightarrow g^{-1}g' \in N \subset M$. Thus $gM = g'M$ and Φ is well-defined.

$aH = bH \Leftrightarrow a^{-1}b \in H$